

X01 MATERIAL RESTRICTIONS | P

Intent : To reduce or eliminate human exposure to building materials known to be hazardous.

Summary : This WELL feature restricts widely known hazardous ingredients in newly installed building materials, specifically asbestos, mercury and lead.

Issue : Historical use of hazardous materials in construction, specifically asbestos, mercury and lead, have presented serious and negative health impacts on humans. Disease caused by exposure to these chemicals, including asbestosis, developmental issues in children¹ and various forms of cancer still affect millions of people.² While these compounds have been restricted or banned in buildings in many countries, they still pose as threats in countries that have not enacted the necessary limitations. As a result, exposure to these hazardous materials should be limited and, if possible, eliminated.

Solutions : Asbestos has been fully or partially banned for buildings in most countries and alternatives are widely available. Lead content in materials that may expose humans to its aspiration and ingestion is also restricted in many national regulations.³ Use of products where lead content is minimized or not added can significantly reduce leaching from pipes into drinking water. Eliminating the use of compact fluorescent light bulbs (CFLs) removes a potential pathway for exposure to mercury.

PART 1 RESTRICT ASBESTOS

For All Spaces:

For newly installed or applied products within the project boundary, the following requirement is met:

- a. The following product categories do not contain over 1,000 ppm of asbestos by weight or area:
 1. Thermal protection, including all insulation (lagging) applied to pipes, fittings, boilers, tanks and ducts.
 2. Acoustic treatments.
 3. Sheathing.
 4. Roofing and siding.
 5. Fire and smoke protection.
 6. Joint protection.
 7. Plaster and gypsum board.
 8. Ceilings.
 9. Resilient flooring.

PART 2 RESTRICT MERCURY

For All Spaces:

The following requirements are met:

- a. Newly installed fluorescent, metal halide and sodium lamps, if present, meet one of the following:
 1. RoHS restrictions.⁴
 2. The following specifications:⁵

Fluorescent Lamp	Maximum Mercury Content
Compact, integral ballast	3.5 mg
Compact, no-integral ballast	3.5 mg
T-5, circular	9 mg
T-5, linear	2.5 mg
T-8, eight-foot	10 mg
T-8, four-foot	3.5 mg
T-8, U-bent	6 mg
High-Pressure Sodium Lamp	Maximum Mercury Content
400 W or less	10 mg
Over 400 W	32 mg

- b. Newly installed fire alarm notification and initiating devices (e.g. strobes, pull stations); environmental, HVAC, occupancy and motion sensors and meters; and relays, thermostats and load break switches meet one of the following:
 1. RoHS restrictions.⁴
 2. Products contain no more than 0.1% (1000 ppm) of mercury by weight.

PART 3 RESTRICT LEAD

For All Spaces:

1: Paints and electronics

The following requirements are met:

- a. Newly installed fire alarms notification and initiating devices (e.g. strobes, pull stations); environmental, HVAC, occupancy and motion sensors and meters; and relays, thermostats and load break switches meet one of the following:
 1. RoHS restrictions.⁴
 2. Products contain no more than 0.1% (1000 ppm) of lead by weight.
- b. Newly installed paints applied as finishes within the project boundary meet at least one of the following criteria:
 1. Paints have a lead concentration of 100 ppm (0.01%) by weight or below.
 2. Paints have no added lead carbonates and lead sulfates.
 3. Paints are deemed free of lead or with no added lead by an ISO 14024-compliant (Type 1) Ecolabel, or a voluntary third-party certification program recognized by the local government where the building is located.
 4. Paints meet Feature X08: Materials Optimization Part 1, Option 1 'Materials Selection' OR Part 2.

2: Drinking water pipes, fittings and solder

Pipes, fixtures, fittings and solder newly installed or applied within the project boundary intended for drinking water distribution and delivery meet at least one of the following:

- a. The product is approved for use with drinking water by a local government authority or by a government-authorized certification body.
- b. The product has a weighted wetted average of 0.25% of lead or less, verified by a third party, or is labeled as ANSI/NSF 372-compliant.

REFERENCES

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2. Furuya S, Chimed-Ochir O, Takahashi K, David A, Takala J. Global Asbestos Disaster. *International Journal of Environmental Research and Public Health*. 2018;15(5).
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5. U.S. Green Building Council. LEED BD+C: Healthcare | v4.1 - LEED v4.1 PBT source reduction - mercury. USGBC. <https://www.usgbc.org/node/12022740>. Published 2019. Accessed Accessed on December 19, 2019, 2019.